



Machine Learning and the Physical World

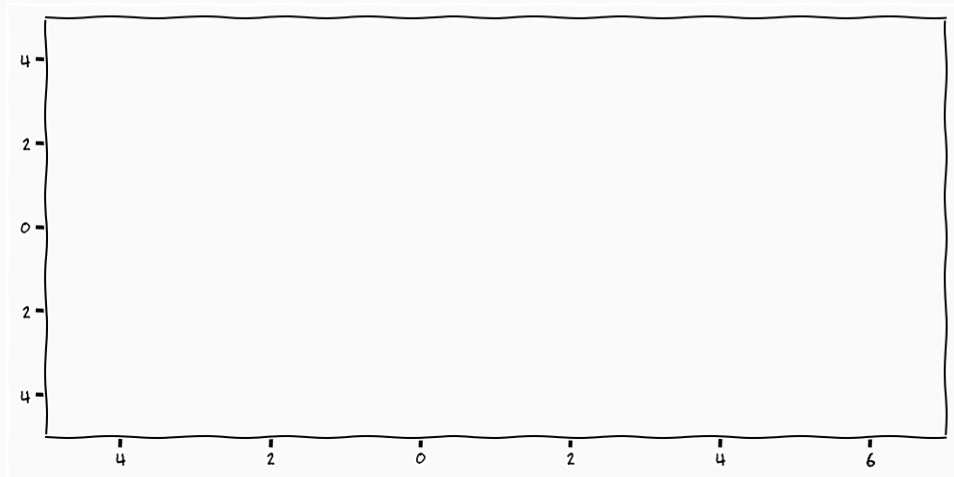
Lecture 5 : Practical Gaussian Processes

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27th of October, 2025

<http://carlhenrik.com>

Functions

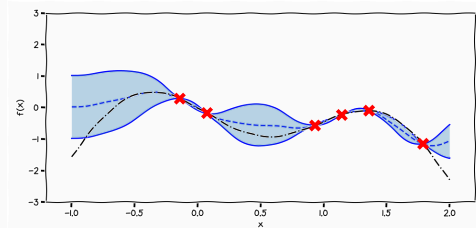
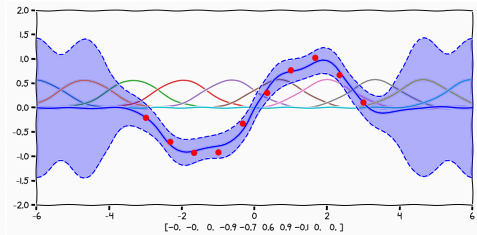


Parametrics vs. Non-parametrics

$$y_i = \underbrace{\mathbf{w}^T}_{\text{random}} \phi(\mathbf{x}_i) + \epsilon_i$$

$$y_i = \underbrace{f_i}_{\text{random}} + \epsilon_i$$

Non-parametrics vs Parametrics



$$p(\mathbf{f}) = \mathcal{N} \left(\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \\ \vdots \end{bmatrix} \middle| \begin{bmatrix} \mu(x_1) \\ \mu(x_2) \\ \vdots \\ \mu(x_N) \\ \vdots \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & k(x_1, x_2) & \dots & k(x_1, x_N) & \dots \\ k(x_2, x_1) & k(x_2, x_2) & \dots & k(x_2, x_N) & \dots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ k(x_N, x_1) & k(x_N, x_2) & \dots & k(x_N, x_N) & \dots \\ \vdots & \vdots & \dots & \vdots & \ddots \end{bmatrix} \right)$$

- Co-variance and mean-function both have parameters

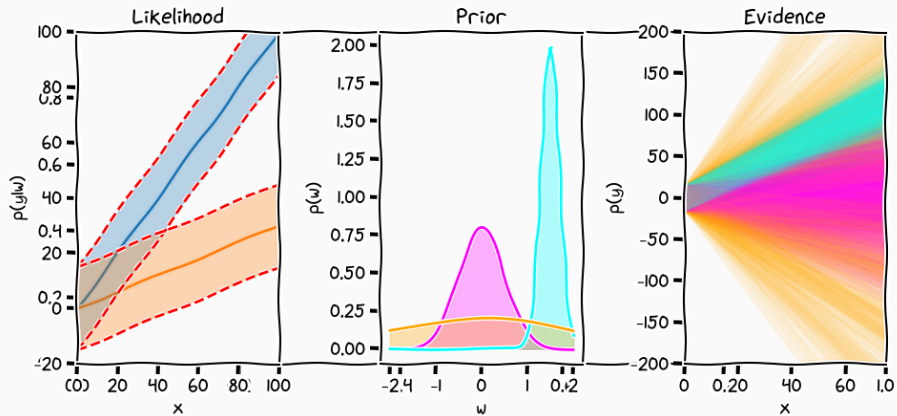
$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{\underbrace{\int p(y \mid \theta)p(\theta)d\theta}_{p(y)}}$$

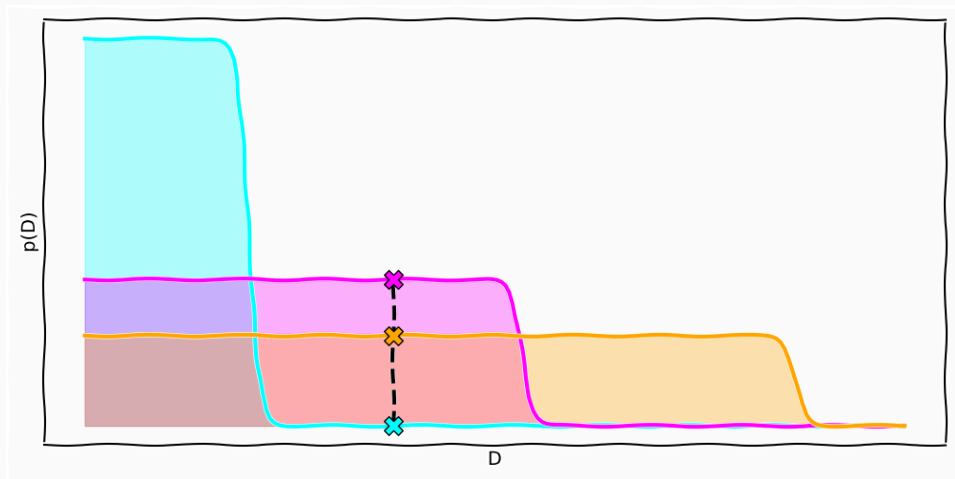
Likelihood How much **evidence** is there in the data for a specific hypothesis

Prior What are my beliefs about different hypothesis

Posterior What is my **updated** belief after having seen data

Evidence What is my belief about the data





$$\boldsymbol{\theta}^* = \operatorname{argmax}_{\boldsymbol{\theta}} p_{\boldsymbol{\theta}}(\mathcal{D})$$

$$\begin{aligned}\log p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) &= \log \int p(\mathbf{y} \mid \mathbf{f}) p(\mathbf{f} \mid \mathbf{X}, \boldsymbol{\theta}) d\mathbf{f} \\ &= -\frac{1}{2} \mathbf{y}^T (k(\mathbf{X}, \mathbf{X}) + \beta^{-1} \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log \det (k(\mathbf{X}, \mathbf{X}) + \beta^{-1} \mathbf{I}) \\ &\quad - \frac{N}{2} \log 2\pi\end{aligned}$$

$$p(y) = \int p(y | f)p(f)df$$

Code

```
import numpy as np
```

```
.....
```

- GPy
- GPFlow
- GPTorch
- GPJax

Numerical Stability

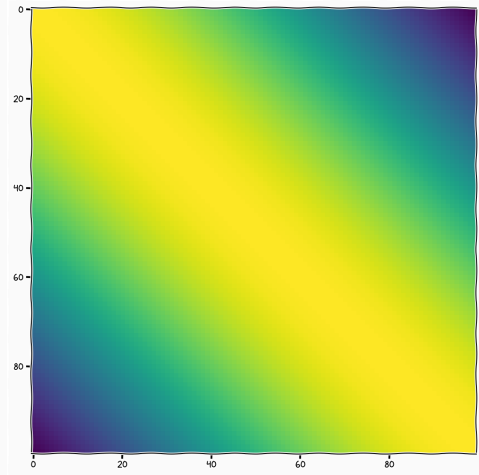
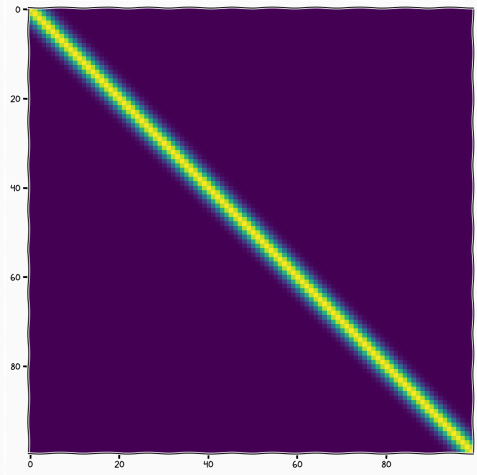
$$\boldsymbol{\theta}^* = \operatorname{argmax}_{\boldsymbol{\theta}} -\mathbf{y}^T \left(k(\mathbf{X}, \mathbf{X}) + \beta^{-1} \mathbf{I} \right)^{-1} \mathbf{y} - \log \det \left(k(\mathbf{X}, \mathbf{X}) + \beta^{-1} \mathbf{I} \right)$$

$$\log \det (k(\mathbf{X}, \mathbf{X}) + \beta^{-1} \mathbf{I})$$

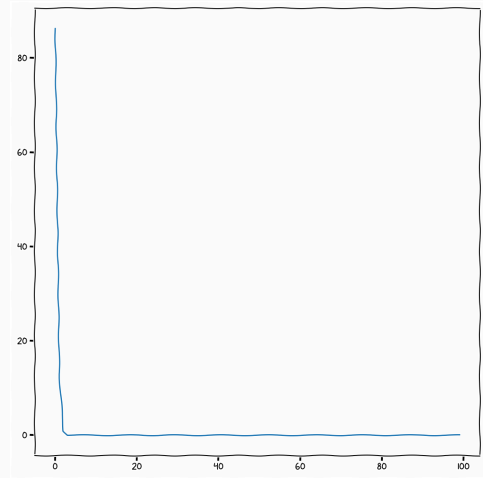
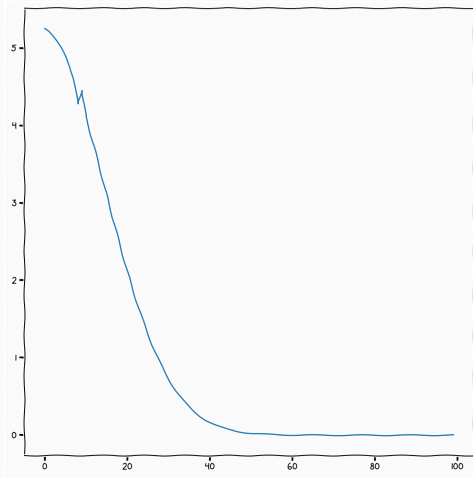
Code

```
np.log(np.linalg.det(K+1/beta*np.eye(K.shape[0])))
```

Co-Variance Matrices



Eigen-decomposition



$$\mathbf{K} = \mathbf{L}\mathbf{L}^T$$

- Factorisation of a *Hermitian* and *Positive-Semi-Definite* Matrix into the product of two lower-triangular matrices

$$\begin{aligned}\log \det \mathbf{K} &= \log \det (\mathbf{L}\mathbf{L}^T) = \\ &\log (\det \mathbf{L})^2 = \\ &\log \left(\prod_i^N \ell_{ii} \right)^2 = 2 \sum_i^N \ell_{ii}\end{aligned}$$

$$\mathbf{y}^T \mathbf{K}^{-1} \mathbf{y}$$

- Matrix inverse have cubic complexity $\mathcal{O}(n^3)$
- Finding the general inverse is numerically tricky
- The matrix is structured and we do not need the explicit matrix

$$\mathbf{y}^T \mathbf{K}^{-1} \mathbf{y} = \mathbf{y}^T (\mathbf{L}\mathbf{L})^T{}^{-1} \mathbf{y}$$

$$\begin{aligned}\mathbf{y}^T \mathbf{K}^{-1} \mathbf{y} &= \mathbf{y}^T (\mathbf{L}\mathbf{L})^T{}^{-1} \mathbf{y} \\ &= \mathbf{y}^T (\mathbf{L}^{-1})^T \mathbf{L}^{-1} \mathbf{y}\end{aligned}$$

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$$\begin{aligned}\mathbf{y}^T \mathbf{K}^{-1} \mathbf{y} &= \mathbf{y}^T (\mathbf{L}\mathbf{L})^T{}^{-1} \mathbf{y} \\ &= \mathbf{y}^T (\mathbf{L}^{-1})^T \mathbf{L}^{-1} \mathbf{y} \\ &= (\mathbf{L}^{-1} \mathbf{y})^T \mathbf{L}^{-1} \mathbf{y} \\ &= \mathbf{z}^T \mathbf{z}.\end{aligned}$$

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$$\mathbf{L}\mathbf{z} = \mathbf{y}$$

$$\begin{aligned}\ell_{1,1}z_1 &= y_1 \\ \ell_{2,1}z_1 + \ell_{2,2}z_2 &= y_2 \\ \vdots & \\ \ell_{n,1}z_1 + \ell_{n,2}z_2 + \dots + \ell_{n,n}z_n &= y_n,\end{aligned}$$

- we can easily solve $z_1 = \frac{y_1}{\ell_{1,1}}$ and $z_2 = \frac{y_1 - \ell_{2,1}z_1}{\ell_{2,2}}$, etc.
- `scipy.linalg.cho_solve`

- A numerical method is an "approximation"

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- Our computers have finite precision

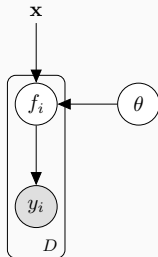
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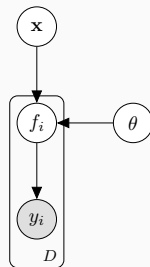
- A numerical method is an "approximation"
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- Even "worse" they have floating finite precision
- Keep the computer in mind when formulating your problem
- There is a "big" forgotten step going from math to code, don't forget your numerical analysis

Intractabilities

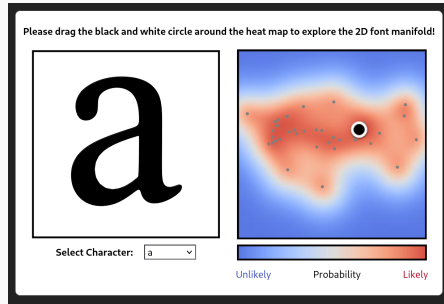
Unsupervised Learning



$$p(y|x) = \int p(y | f)p(f)df$$

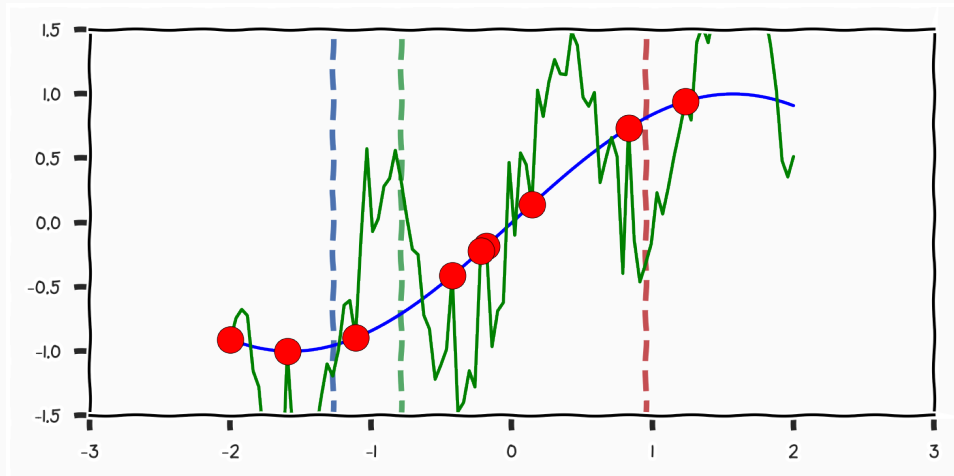


$$p(y) = \int p(y | f, x)p(f | x)p(x)dfdx$$

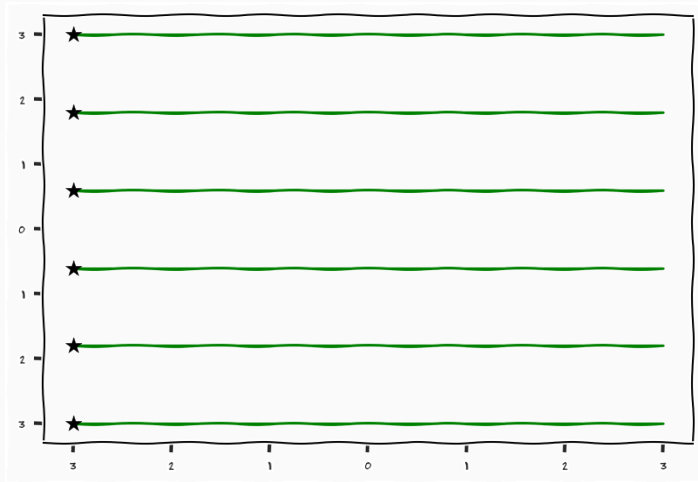


NDF Campbell et al. (July 2014). “Learning a manifold of fonts”. In: *ACM Transactions on Graphics (TOG)* 33.4, p. 91

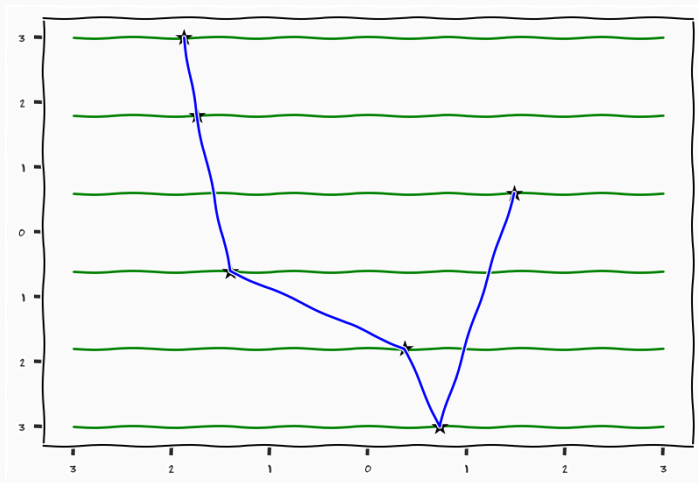
Functions



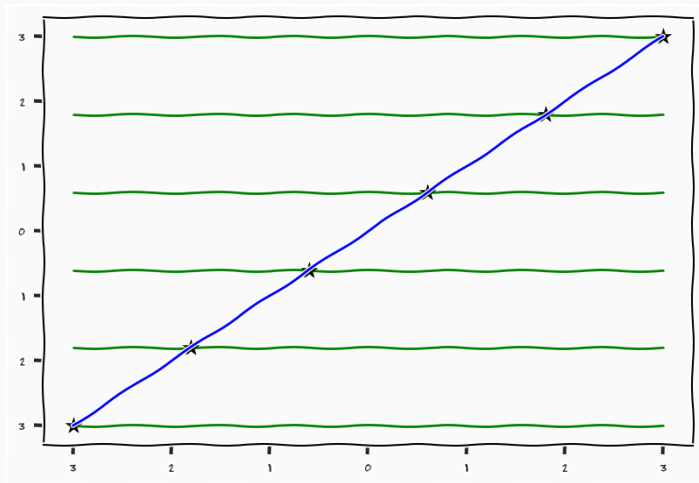
Unsupervised Learning



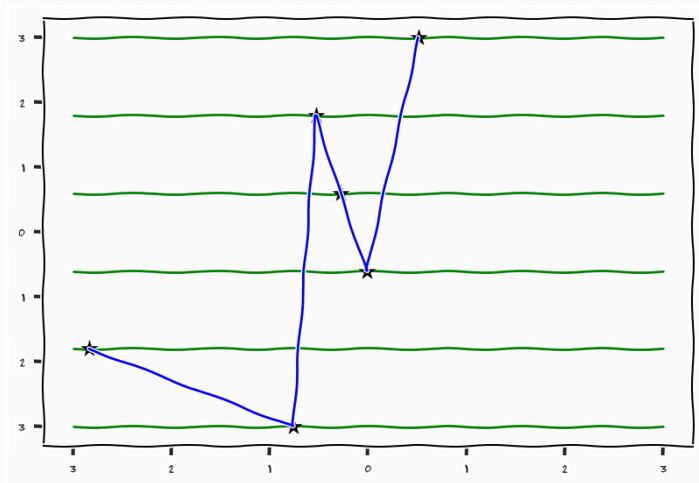
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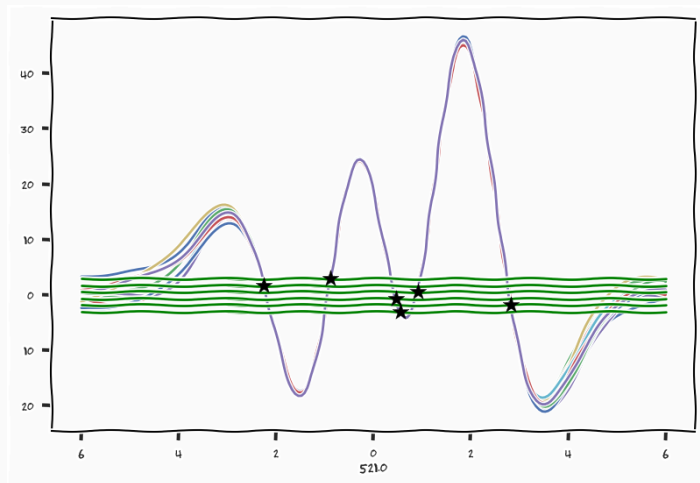
Unsupervised Learning



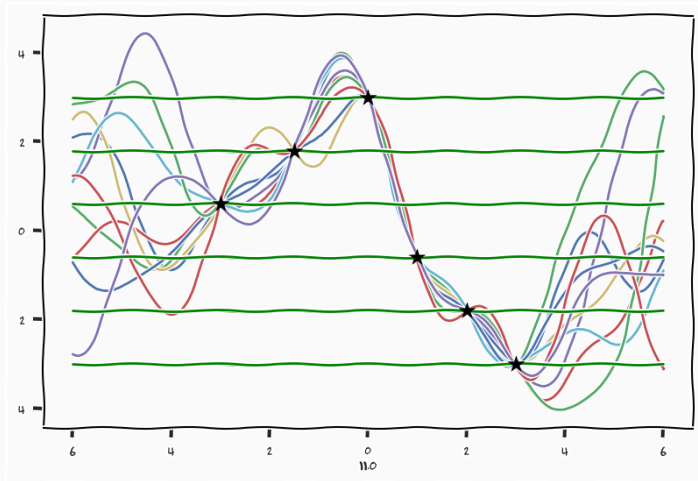
Unsupervised Learning



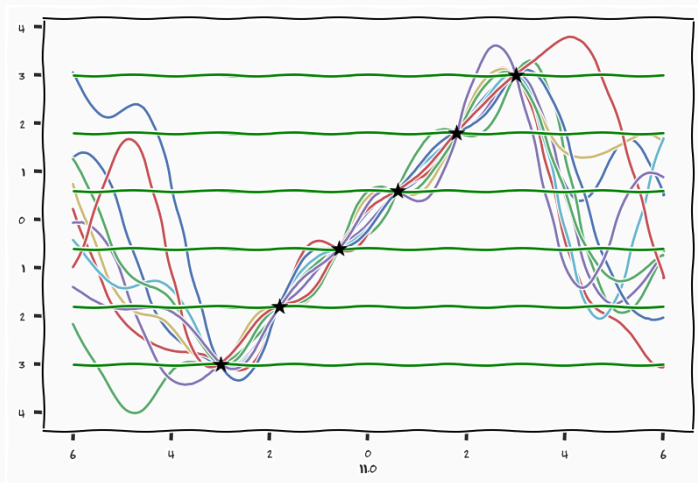
Unsupervised Learning



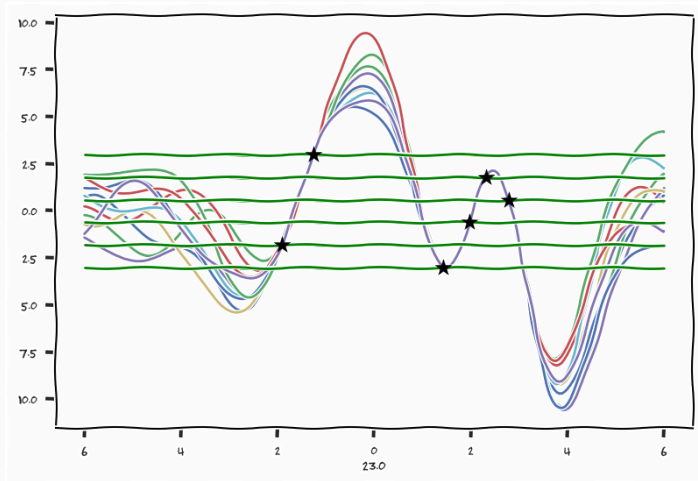
Unsupervised Learning



Unsupervised Learning



Unsupervised Learning



Regression there are infinite number of possible functions that connects the data equally well. A GP provides a measure over these solutions that makes the problem "well-posed".

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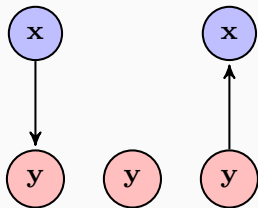
Unsupervised Learning there are infinite number of possible combinations of input locations and functions that generate the data equally well. A GP and a latent space prior jointly provides a measure over these solutions to make the problem "well-posed"

$$p(y) = \int p(y | f)p(f | x)p(x)dfdx$$

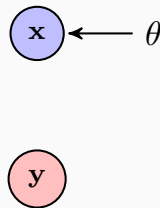
- This integral is analytically intractable

Approximate Inference

$$p(y) = \int p(y \mid x)p(x)dx$$



$$p(y) = \int_x p(y|x)p(x) = \frac{p(y|x)p(x)}{p(x|y)}$$



$$q_{\theta}(x) \approx p(x|y)$$

$$p(y)$$

$$\log p(y)$$

$$\log p(y) = \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx$$

$$\begin{aligned}\log p(y) &= \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx \\ &= \int q(x) \log p(y) dx + \int q(x) \log \frac{p(x|y)}{p(x|y)} dx\end{aligned}$$

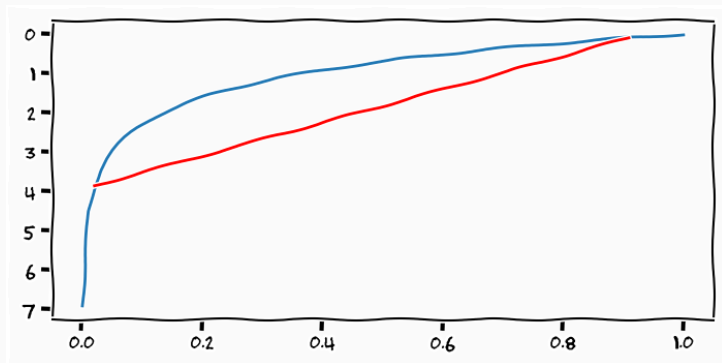
$$\begin{aligned}\log p(y) &= \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx \\ &= \int q(x) \log p(y) dx + \int q(x) \log \frac{p(x|y)}{p(x|y)} dx \\ &= \int q(x) \log \frac{p(x|y)p(y)}{p(x|y)} dx\end{aligned}$$

$$\begin{aligned}\log p(y) &= \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx \\ &= \int q(x) \log p(y) dx + \int q(x) \log \frac{p(x|y)}{p(x|y)} dx \\ &= \int q(x) \log \frac{p(x|y)p(y)}{p(x|y)} dx = \int q(x) \log \frac{p(x, y)}{p(x|y)} dx\end{aligned}$$

$$\begin{aligned}\log p(y) &= \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx \\&= \int q(x) \log p(y) dx + \int q(x) \log \frac{p(x|y)}{p(x|y)} dx \\&= \int q(x) \log \frac{p(x|y)p(y)}{p(x|y)} dx = \int q(x) \log \frac{p(x, y)}{p(x|y)} dx \\&= \int q(x) \log \frac{q(x)}{q(x)} dx + \int q(x) \log p(x, y) dx + \int q(x) \log \frac{1}{p(x|y)} dx\end{aligned}$$

$$\begin{aligned}\log p(y) &= \log p(y) + \int \log \frac{p(x|y)}{p(x|y)} dx \\&= \int q(x) \log p(y) dx + \int q(x) \log \frac{p(x|y)}{p(x|y)} dx \\&= \int q(x) \log \frac{p(x|y)p(y)}{p(x|y)} dx = \int q(x) \log \frac{p(x, y)}{p(x|y)} dx \\&= \int q(x) \log \frac{q(x)}{q(x)} dx + \int q(x) \log p(x, y) dx + \int q(x) \log \frac{1}{p(x|y)} dx \\&= \int q(x) \log \frac{1}{q(x)} dx + \int q(x) \log p(x, y) dx + \int q(x) \log \frac{q(x)}{p(x|y)} dx\end{aligned}$$

Jensen Inequality



$$f\left(\int g \, dx\right) \leq \int f \circ g \, dx,$$

The "posterior" term

$$\int q(x) \log \frac{q(x)}{p(x|y)} dx$$

The "posterior" term

$$\int q(x) \log \frac{q(x)}{p(x|y)} dx = - \int q(x) \log \frac{p(x|y)}{q(x)} dx$$

The "posterior" term

$$\begin{aligned}\int q(x) \log \frac{q(x)}{p(x|y)} dx &= - \int q(x) \log \frac{p(x|y)}{q(x)} dx \\ &\geq \log \int p(x|y) dx \\ &= \log 1 = 0\end{aligned}$$

The "posterior" term

$$\int q(x) \log \frac{q(x)}{p(x|y)} dx$$

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$$\int q(x) \log \frac{q(x)}{p(x|y)} dx = \{\text{Lets assume that } q(x) = p(x|y)\}$$

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$$\begin{aligned}\int q(x) \log \frac{q(x)}{p(x|y)} dx &= \{\text{Lets assume that } q(x) = p(x|y)\} \\ &= \int p(x|y) \log \underbrace{\frac{p(x|y)}{p(x|y)}}_{=1} dx\end{aligned}$$

The "posterior" term

$$\begin{aligned}\int q(x) \log \frac{q(x)}{p(x|y)} dx &= \{\text{Lets assume that } q(x) = p(x|y)\} \\ &= \int p(x|y) \log \underbrace{\frac{p(x|y)}{p(x|y)}}_{=1} dx \\ &= 0\end{aligned}$$

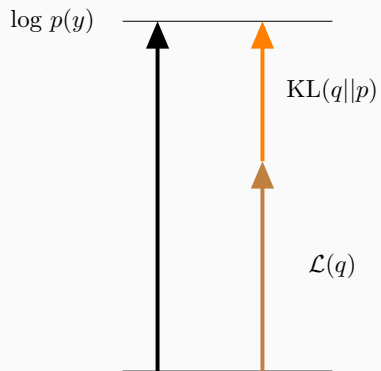
$$\text{KL}(q(x)||p(x|y)) = \int q(x) \log \frac{q(x)}{p(x|y)} dx$$

- $\text{KL}(q(x)||p(x|y)) \geq 0$
- $\text{KL}(q(x)||p(x|y)) = 0 \Leftrightarrow q(x) = p(x|y)$
- Measure of divergence between distributions
- Not a metric (not symmetric)

$$\begin{aligned}\log p(y) &= \int q(x) \log \frac{1}{q(x)} dx + \int q(x) \log p(x, y) dx + \int q(x) \log \frac{q(x)}{p(x|y)} dx \\ &\geq - \int q(x) \log q(x) dx + \int q(x) \log p(x, y) dx\end{aligned}$$

- The Evidence Lower BOnd
- Tight if $q(x) = p(x|y)$

Deterministic Approximation



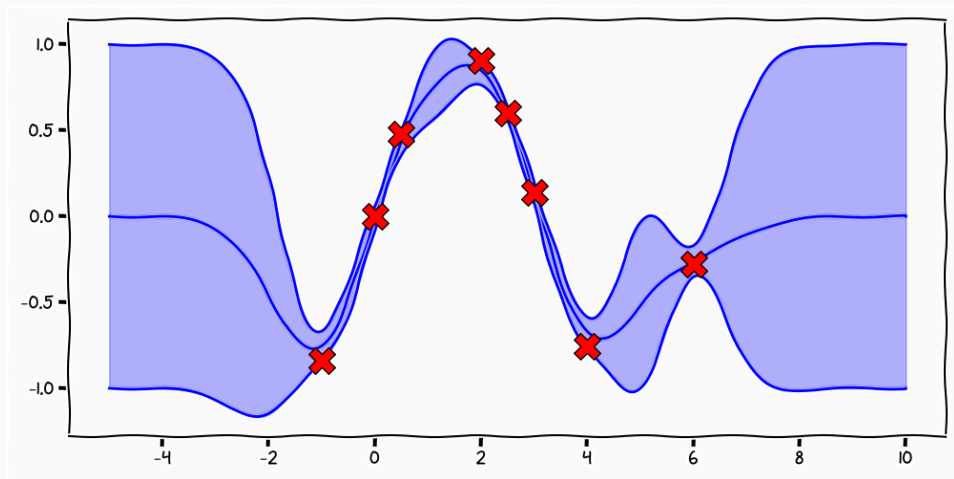
$$\begin{aligned}\log p(y) &\geq - \int q(x) \log q(x) dx + \int q(x) \log p(x, y) dx \\ &= \mathbb{E}_{q(x)} [\log p(x, y)] - H(q(x)) = \mathcal{L}(q(x))\end{aligned}$$

- if we maximise the ELBO we,
 - find an approximate posterior
 - lower bound the marginal likelihood
- *maximising* $p(y)$ **is** learning
- finding $q(x) \approx p(x|y)$ **is** prediction

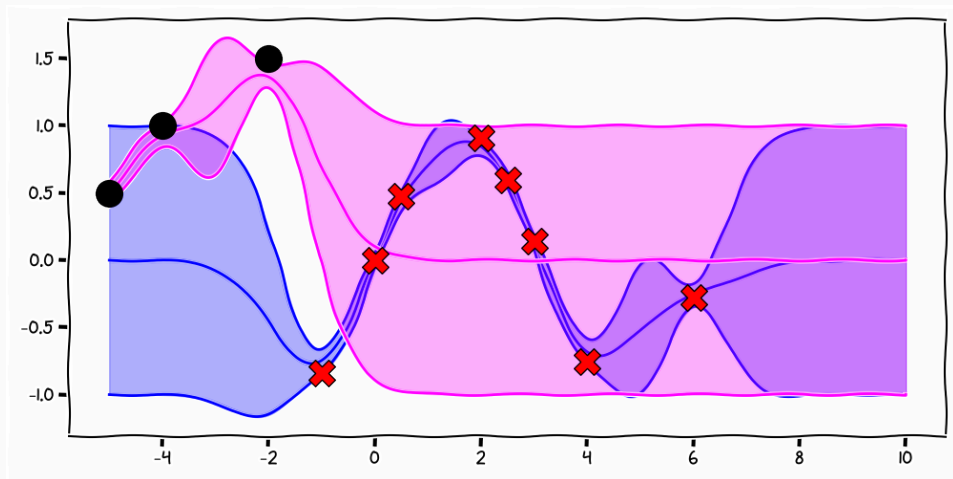
$$\mathcal{L}(q(x)) = \mathbb{E}_{q(x)} [\log p(x, y)] - H(q(x))$$

- We have to be able to compute an expectation over the joint distribution
- The second term should be trivial

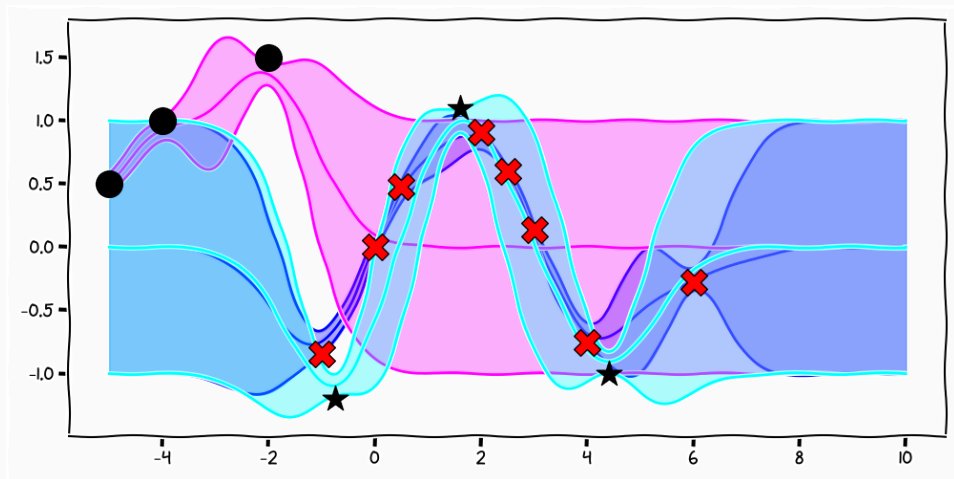
Gaussian Processes Q



Gaussian Processes Q



Gaussian Processes Q



$$p(f, u \mid x, z)$$

- Add another set of samples from the same prior
- Conditional distribution

¹Titsias et al., [2010](#)

$$p(f, u \mid x, z) = p(f \mid u, x, z)p(u \mid z)$$

- Add another set of samples from the same prior
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¹Titsias et al., [2010](#)

$$\begin{aligned} p(f, u \mid x, z) &= p(f \mid u, x, z) p(u \mid z) \\ &= \mathcal{N}(f \mid K_{fu} K_{uu}^{-1} u, K_{ff} - K_{fu} K_{uu}^{-1} K_{uf}) \mathcal{N}(u \mid \mathbf{0}, K_{uu}) \end{aligned}$$

- Add another set of samples from the same prior
- Conditional distribution

¹Titsias et al., [2010](#)

$$p(y, f, u, x \mid z) = p(y \mid f)p(f \mid u, x)p(u \mid z)p(x)$$

- we have done nothing to the model, just project an additional set of marginals from the GP
- *however* we will now **interpret** u and z not as **random** variables but **variational** parameters
- i.e. the variational distribution $q(\cdot)$ is parametrised by these

- Variational distributions are approximations to intractable posteriors,

$$q(u) \approx p(u \mid y, x, z, f)$$

$$q(f) \approx p(f \mid u, x, z, y)$$

$$q(x) \approx p(x \mid y)$$

- Variational distributions are approximations to intractable posteriors,

$$q(u) \approx p(u \mid y, x, z, f)$$

$$q(f) \approx p(f \mid u, x, z, y)$$

$$q(x) \approx p(x \mid y)$$

- Bound is **tight** if u completely represents f i.e. u is sufficient statistics for f

$$q(f) \approx p(f \mid u, x, z, y) = p(f \mid u, x, z)$$

$$\mathcal{L} = \int_{x,f,u} q(f) q(u) q(x) \log \frac{p(y, f, u \mid x, z) p(x)}{q(f) q(u) q(x)}$$

$$\begin{aligned}\mathcal{L} &= \int_{x,f,u} q(f)q(u)q(x) \log \frac{p(y, f, u \mid x, z)p(x)}{q(f)q(u)q(x)} \\ &= \int_{x,f,u} q(f)q(u)q(x) \log \frac{p(y \mid f)p(f \mid u, x, z)p(u \mid z)p(x)}{q(f)q(u)q(x)}\end{aligned}$$

- Assume that u is sufficient statistics of f

$$q(f) = p(f \mid u, x, z)$$

$$\tilde{\mathcal{L}} = \int_{x,f,u} q(f) q(u) q(x) \log \frac{p(y | f) p(f | u, x, z) p(u | z) p(x)}{q(f) q(u) q(x)}$$

$$\begin{aligned}\tilde{\mathcal{L}} &= \int_{x,f,u} q(f)q(u)q(x) \log \frac{p(y|f)p(f|u,x,z)p(u|z)p(x)}{q(f)q(u)q(x)} \\ &= \int_{x,f,u} p(f|u,x,z)q(u)q(x) \log \frac{p(y|f)p(f|u,x,z)p(u|z)p(x)}{p(f|u,x,z)q(u)q(x)}\end{aligned}$$

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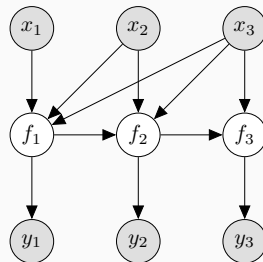
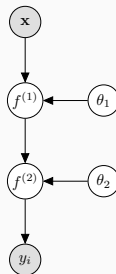
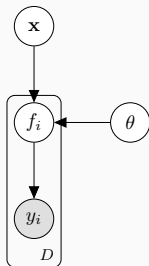
$$\begin{aligned}\tilde{\mathcal{L}} &= \int_{x,f,u} q(f)q(u)q(x) \log \frac{p(y|f)p(f|u,x,z)p(u|z)p(x)}{q(f)q(u)q(x)} \\ &= \int_{x,f,u} p(f|u,x,z)q(u)q(x) \log \frac{p(y|f)\color{red}{p(f|u,x,z)}p(u|z)p(x)}{\color{red}{p(f|u,x,z)}q(u)q(x)} \\ &= \int_{x,f,u} p(f|u,x,z)q(u)q(x) \log \frac{p(y|f)p(u|z)p(x)}{q(u)q(x)}\end{aligned}$$

$$\begin{aligned}\tilde{\mathcal{L}} &= \int_{x,f,u} q(f)q(u)q(x) \log \frac{p(y|f)p(f|u,x,z)p(u|z)p(x)}{q(f)q(u)q(x)} \\ &= \int_{x,f,u} p(f|u,x,z)q(u)q(x) \log \frac{p(y|f)\textcolor{red}{p(f|u,x,z)}p(u|z)p(x)}{\textcolor{red}{p(f|u,x,z)}q(u)q(x)} \\ &= \int_{x,f,u} p(f|u,x,z)q(u)q(x) \log \frac{p(y|f)p(u|z)p(x)}{q(u)q(x)} \\ &= \mathbb{E}_{p(f|u,x,z)}[p(y|f)] - \text{KL}(q(u) \parallel p(u|z)) - \text{KL}(q(x) \parallel p(x))\end{aligned}$$

$$\mathcal{L} = \mathbb{E}_{p(f|u,x,z)}[p(y | f)] - \text{KL}(q(u) \parallel p(u | z)) - \text{KL}(q(x) \parallel p(x))$$

- Expectation tractable (for some co-variances) Titsias et al., 2010
- Stochastic inference Hensman et al., 2013
- Importantly $p(x)$ only appears in $\text{KL}(\cdot \parallel \cdot)$ term!

Interesting Models



Summary

- Hopefully this gave you a flavour of the "practical" part of working with probabilistic models
- You are not expected to know this, but having it in the back of your mind
- Remember the no-free lunch, any result is relative to the assumptions that you put in
- Computations and implementations makes up a huge part of your assumptions

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